

1 When Is Consumption Growth Declining in m ?

Define $\chi(\mu) = \log c(\exp(\mu))$.

Figure 4 depicts the expected consumption growth factor as a strictly declining function of the cash-on-hand ratio. To investigate this, define

$$\begin{aligned}
 v(\mu_t) &\equiv \log \Phi_{t+1} c(\mathcal{R}_{t+1} a(m_t) + \xi_{t+1}) / c(m_t) \\
 &= \log \Phi_{t+1} + \log c(\mathcal{R}_{t+1} a(m_t) + \xi_{t+1}) - \log c(m_t) \\
 &= \log \Phi_{t+1} + \chi(\mu_{t+1}) - \chi(\mu_t) \\
 &\approx \log \Phi_{t+1} + \chi(\mu_t) + \chi'(\mu_t)(\mu_{t+1} - \mu_t) - \chi(\mu_t) \\
 &\approx \log \Phi_{t+1} + \chi'(\mu_t)(\mu_{t+1} - \mu_t)
 \end{aligned} \tag{1}$$

and the proposition in which we are interested is

$$(d/d\mu_t) \mathbb{E}_t \underbrace{[v(\mu_t)]}_{\equiv v_{t+1}} < 0$$

or differentiating through the expectations operator, what we want is

$$\mathbb{E}_t [\chi'(\mu_t)(d\mu_{t+1}/d\mu_t - 1) + \chi''(\mu_t)(\mu_{t+1} - \mu_t)] < 0 \tag{2}$$

but in the vicinity of $\mu_{t+1} - \mu_t = 0$ this reduces to

$$\begin{aligned}
 \mathbb{E}_t [\chi'(\mu_t)(d\mu_{t+1}/d\mu_t - 1)] &< 0 \\
 \mathbb{E}_t [d\mu_{t+1}/d\mu_t] &< 1
 \end{aligned} \tag{3}$$

but

$$\begin{aligned}
 \mu_{t+1} &= \log (\mathcal{R}_{t+1} a(m_t) + \xi_{t+1} \Psi_{t+1}) \\
 &= \log (\mathcal{R}_{t+1} a(m_t) (1 + \xi_{t+1} (\Psi_{t+1})^2 \Phi / R)) \\
 &= \log (\mathcal{R}_{t+1} a(m_t)) + \log (1 + \xi_{t+1} (\Psi_{t+1})^2 \Phi / R)
 \end{aligned} \tag{4}$$

$$\begin{aligned}
 \mathbb{E}_t [\log c_{t+1}] &< \log c'(m_t) - (\log R + \log(1 - c'(m_t))) \\
 \mathbb{E}_t [\log (c'(m_t) + c''(m_t)(m_{t+1} - m_t))] &< \log c'(m_t) - (\log R + \log(1 - c'(m_t))) \\
 \mathbb{E}_t [\log (c'(m_t)(1 + c''(m_t)(m_{t+1} - m_t)/c'(m_t)))] &< \log c'(m_t) - (\log R + \log(1 - c'(m_t))) \\
 \log c'(m_t) + \mathbb{E}_t [(\log(1 + c''(m_t)(m_{t+1} - m_t)/c'(m_t)))] &< \log c'(m_t) - (\log R + \log(1 - c'(m_t))) \\
 \mathbb{E}_t [(\log(1 + c''(m_t)(m_{t+1} - m_t)/c'(m_t)))] &< -(\log R + \log(1 - c'(m_t)))
 \end{aligned} \tag{5}$$

Now consider this near the point where $\mathbb{E}_t[m_{t+1}] = m_t$:

$$\begin{aligned}
 0 &< c'(m_t) - \log R \\
 0 &< c'(m_t) - r \\
 r &< r - \rho^{-1}(r - \tau) < c'(m_t) \\
 \rho^{-1}(r - \tau) &< c'(m_t)
 \end{aligned} \tag{6}$$

Henceforth indicating appropriate arguments by the corresponding subscript (e.g.

$c'_{t+1} \equiv c'(m_{t+1})$), since $\Phi_{t+1}\mathcal{R}_{t+1} = R$, the portion of the LHS of equation (6) in brackets can be manipulated to yield

$$\begin{aligned} c_t \Upsilon'_{t+1} &= c'_{t+1} a'_t R - c'_t \Phi_{t+1} c_{t+1} / c_t \\ &= c'_{t+1} a'_t R - c'_t \Upsilon_{t+1}. \end{aligned}$$

Now differentiate the Euler equation with respect to m_t :

$$\begin{aligned} 1 &= R\beta \mathbb{E}_t[\Upsilon_{t+1}^{-\rho}] \\ 0 &= \mathbb{E}_t[\Upsilon_{t+1}^{-\rho-1} \Upsilon'_{t+1}] \\ &= \mathbb{E}_t[\Upsilon_{t+1}^{-\rho-1}] \mathbb{E}_t[\Upsilon'_{t+1}] + \text{cov}_t(\Upsilon_{t+1}^{-\rho-1}, \Upsilon'_{t+1}) \\ \mathbb{E}_t[\Upsilon'_{t+1}] &= -\text{cov}_t(\Upsilon_{t+1}^{-\rho-1}, \Upsilon'_{t+1}) / \mathbb{E}_t[\Upsilon_{t+1}^{-\rho-1}] \end{aligned} \tag{7}$$

but since $\Upsilon_{t+1} > 0$ we can see from (7) that (6) is equivalent to

$$\text{cov}_t(\Upsilon_{t+1}^{-\rho-1}, \Upsilon'_{t+1}) > 0$$

which, using (7), will be true if

$$\text{cov}_t(\Upsilon_{t+1}^{-\rho-1}, c'_{t+1} a'_t R - c'_t \Upsilon_{t+1}) > 0$$

which in turn will be true if both

$$\text{cov}_t(\Upsilon_{t+1}^{-\rho-1}, c'_{t+1}) > 0$$

and

$$\text{cov}_t(\Upsilon_{t+1}^{-\rho-1}, \Upsilon_{t+1}) < 0.$$

The latter proposition is obviously true under our assumption $\rho > 1$. The former will be true if

$$\text{cov}_t((\Phi \Psi_{t+1} c(m_{t+1}))^{-\rho-1}, c'(m_{t+1})) > 0.$$

The two shocks cause two kinds of variation in m_{t+1} . Variations due to ξ_{t+1} satisfy the proposition, since a higher draw of ξ both reduces $c_{t+1}^{-\rho-1}$ and reduces the marginal propensity to consume. However, permanent shocks have conflicting effects. On the one hand, a higher draw of Ψ_{t+1} will reduce m_{t+1} , thus increasing both $c_{t+1}^{-\rho-1}$ and c'_{t+1} . On the other hand, the $c_{t+1}^{-\rho-1}$ term is multiplied by $\Phi \Psi_{t+1}$, so the effect of a higher Ψ_{t+1} could be to decrease the first term in the covariance, leading to a negative covariance with the second term. (Analogously, a lower permanent shock Ψ_{t+1} can also lead a negative correlation.)